



University
of Glasgow

Power Electronics

Switched Mode DC-DC Converters I:

Boost Converter, Buck-Boost Converter

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Please read **Chapter 7** in the textbook

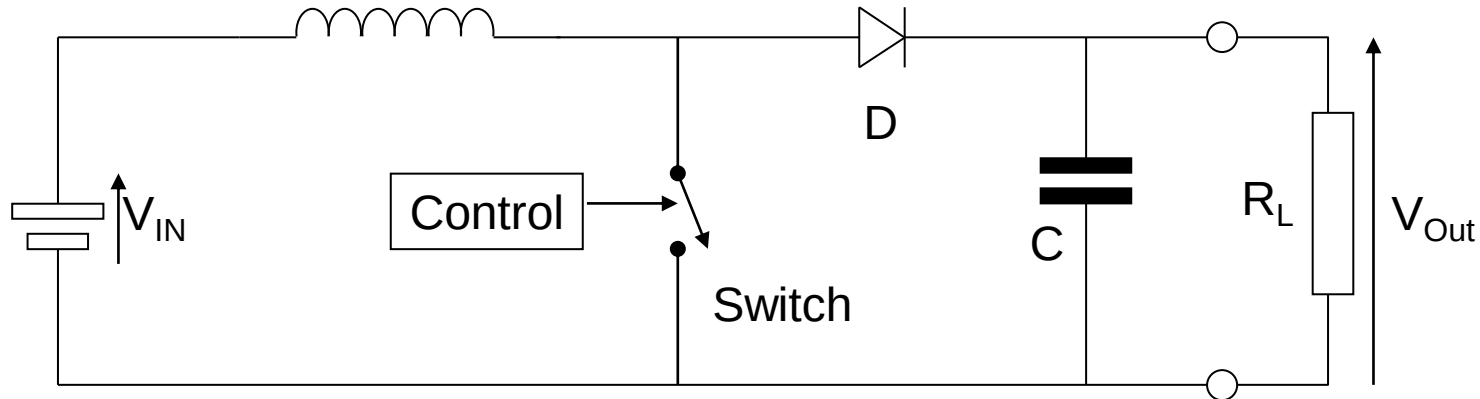
POWER ELECTRONICS

Converters, Applications, and Design

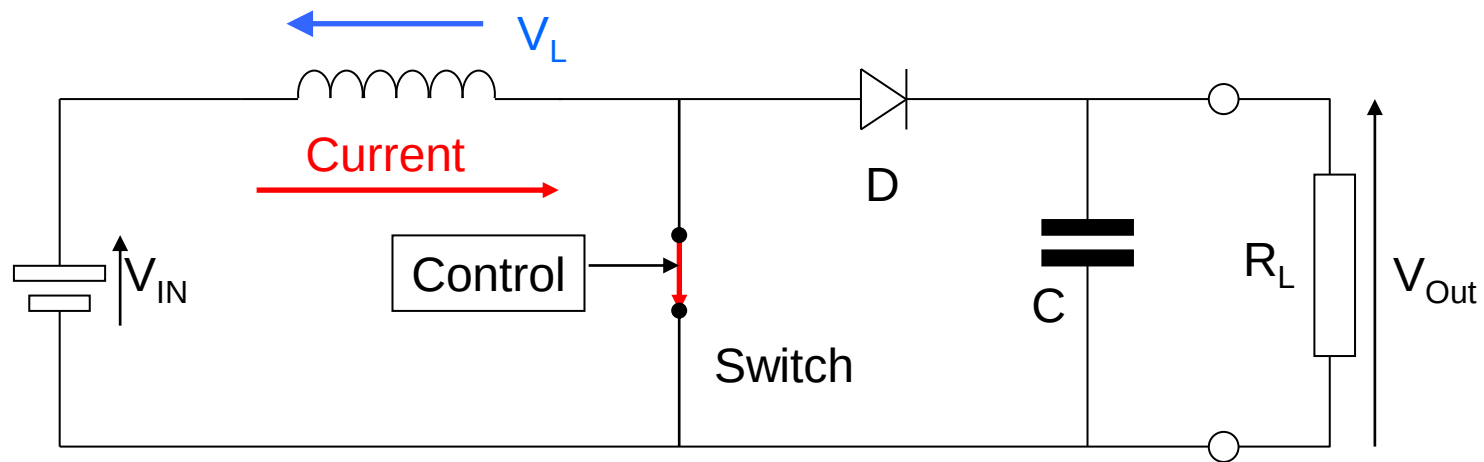
The Boost Converter

- How it works?
- Important waveforms
- Expression for output voltage
 - Based on average voltage of inductor
 - Based on change in current of inductor
- Numerical Problems

The boost (step-up) converter. ($V_{\text{Out}} > V_{\text{In}}$)

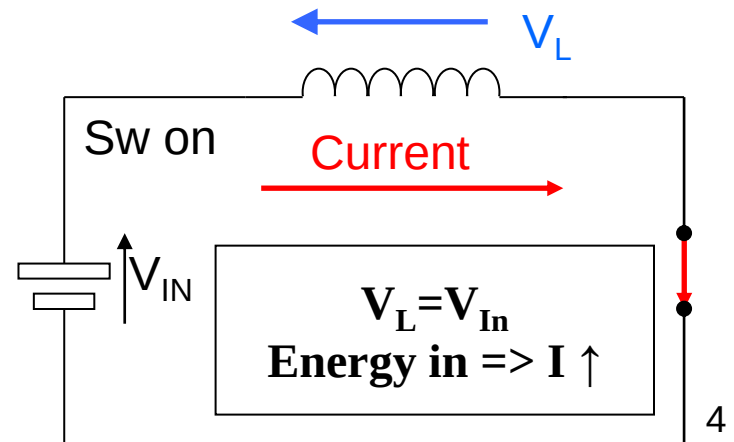


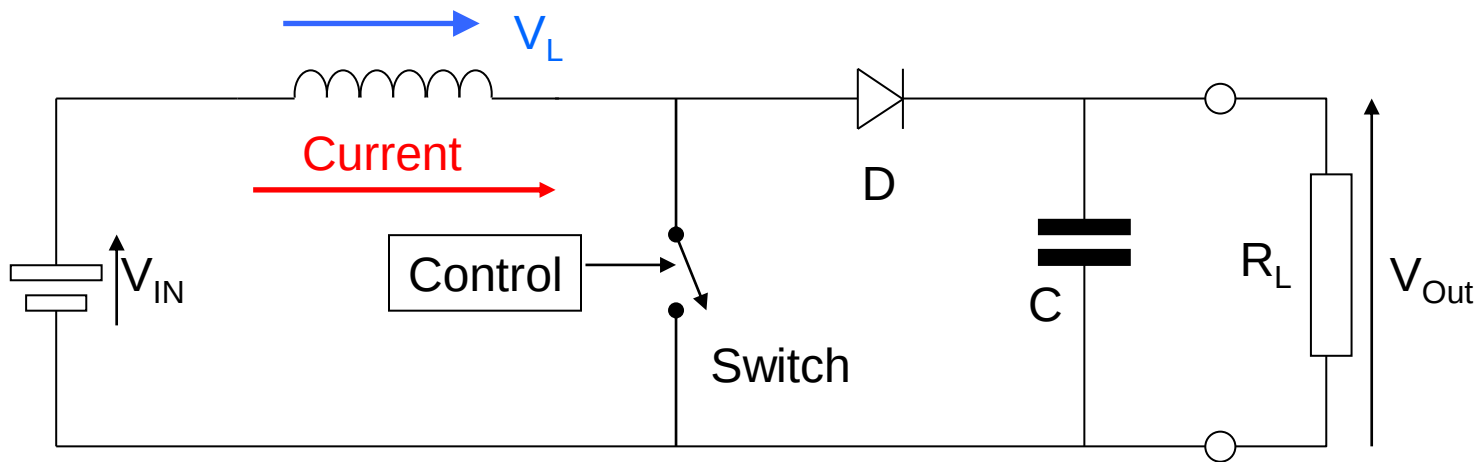
- The boost converter is used to increase the output voltage (V_{Out}) above the input voltage (V_{In}).
- As in the case of the buck converter we will assume the circuit operates in **continuous mode**, i.e., current is always flowing in the inductor.
- If we disregard the switch for a moment and examine the circuit, we see that in the steady state the output voltage is connected to the input voltage via an inductor (assume zero resistance) and a diode. As with the buck, ignore the voltage drop across the diode.



- Now consider the circuit when the switch is closed.
- Any voltage on capacitor C will ensure the diode is reverse biased and therefore the output circuit is “disconnected” from the input. [Because the diode’s anode is connected to ground via the switch.]
- The inductor is connected across the supply (V_{in}) and hence current will start to build through it with the energy being stored in the magnetic field. During this phase, $V_L = V_{in}$.

Switch ON: Charging the inductor !!!



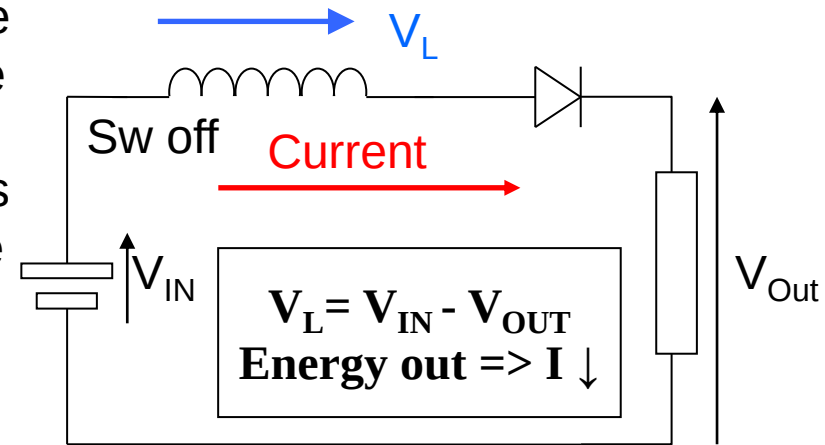


Switch OFF: Discharging the inductor !!!

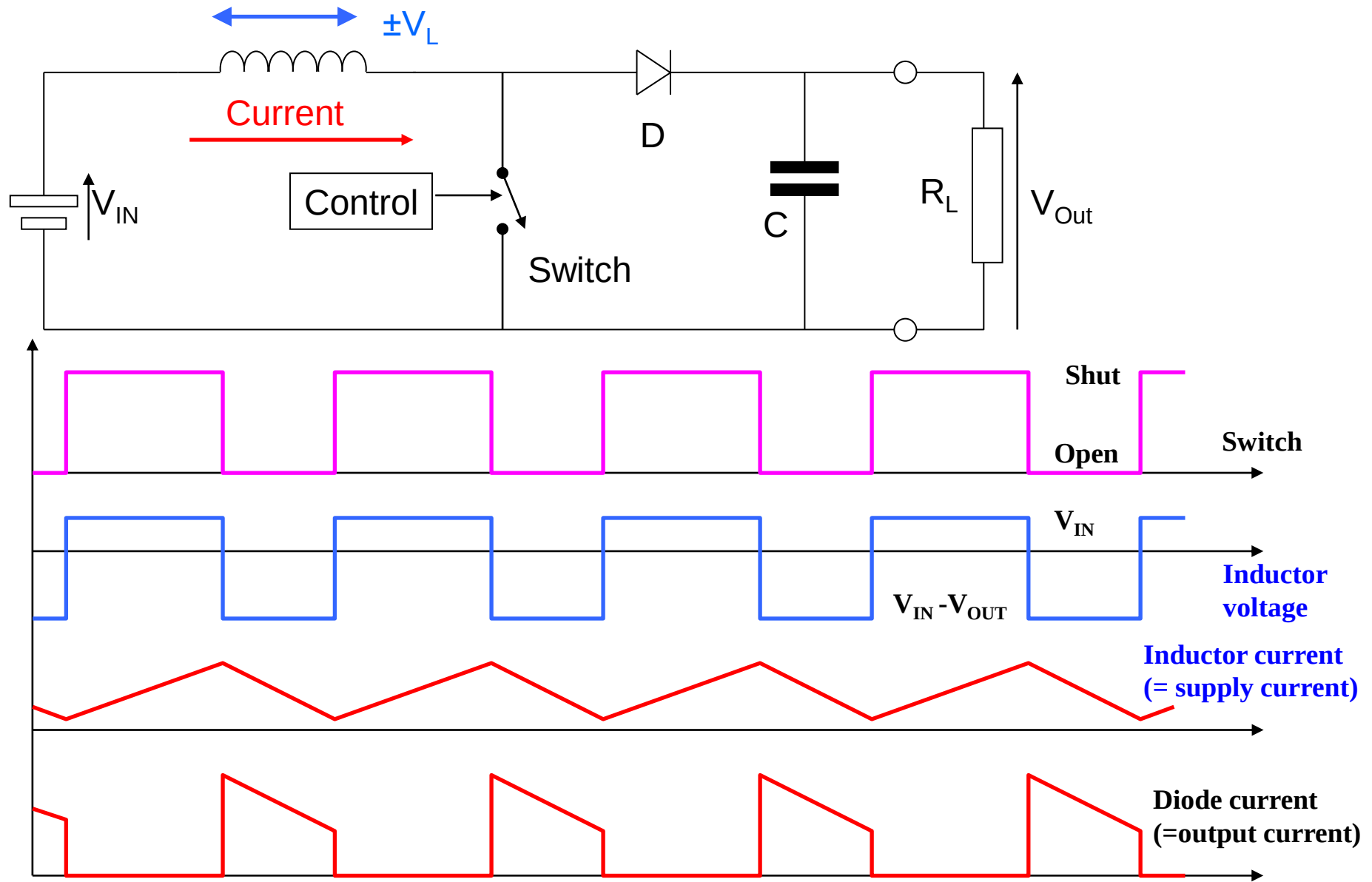
When the switch is opened, the inductor wants to maintain the current flow and with the supply side of the inductor still held at V_{IN} , the potential at the opposite side of the inductor rises until the diode starts to conduct. The inductor voltage during this phase is the difference between the input voltage (V_{IN}) and the output voltage (V_{OUT}).

$$\text{So } V_L = V_{IN} - V_{OUT}$$

As energy is transferred from the inductor's magnetic field so the current flowing through the inductor and diode falls.



Waveforms for the boost converter



Note that current always flows in the inductor (continuous mode again)

Boost converter Input/Output Voltage Relation

The integral of inductor voltage is equal to the change in the inductor flux linkage.

The net change of flux linking the inductor over one time period of repetition is zero, which is a necessary condition for steady-state operation.

$$\int_0^T V_L dt = 0$$

$$\int_0^{t_{on}} V_{IN} dt + \int_{t_{on}}^T (V_{IN} - V_{OUT}) dt = 0$$

$$V_{IN}t_{on} + V_{IN}T - V_{IN}t_{on} - V_{OUT}T + V_{OUT}t_{on} = 0$$

$$(V_{IN} - V_{OUT})T = -V_{OUT}t_{on}$$

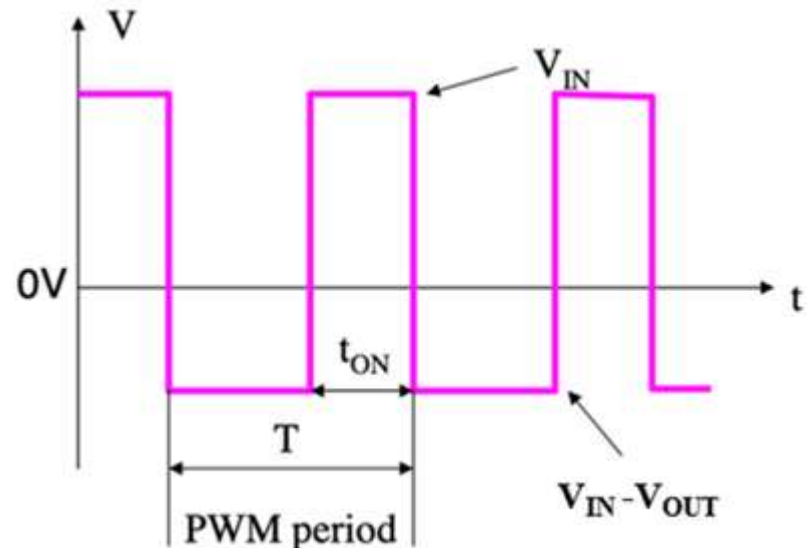
$$(V_{IN} - V_{OUT}) = -V_{OUT} \frac{t_{on}}{T}$$

$$V_{IN} - V_{OUT} = -V_{OUT}D$$

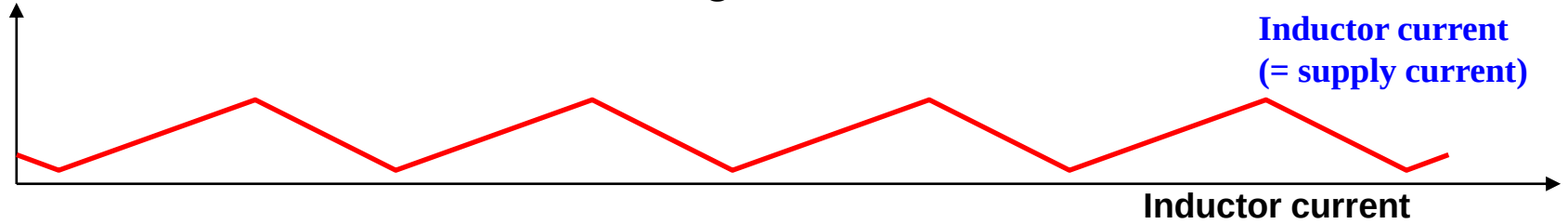
$$V_{IN} = V_{OUT} - V_{OUT}D$$

$$V_{IN} = V_{OUT}(1 - D)$$

$$V_{IN} = V_{OUT}(1 - D) \quad \Rightarrow \quad V_{OUT} = \frac{V_{IN}}{1 - D}$$



Boost converter Inductor Current Change

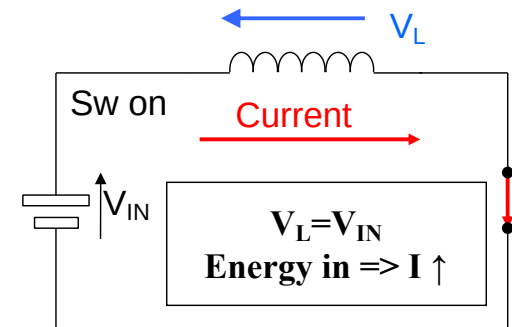


Computing the change in current using following expression:

$$v_L = L \frac{di_L}{dt} \Rightarrow \frac{di_L}{dt} = \frac{v_L}{L}$$

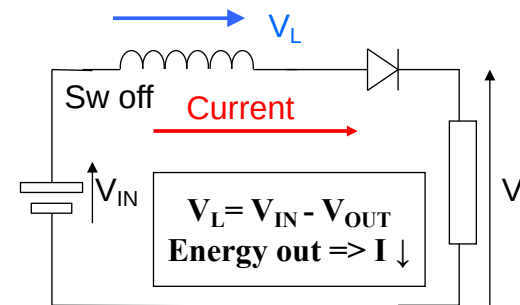
When switch is closed:

$$\frac{(\Delta i_L)_{switch\ closed}}{t_{on}} = \frac{V_{IN}}{L} \Rightarrow (\Delta i_L)_{switch\ closed} = \frac{V_{IN}}{L} \times t_{on}$$

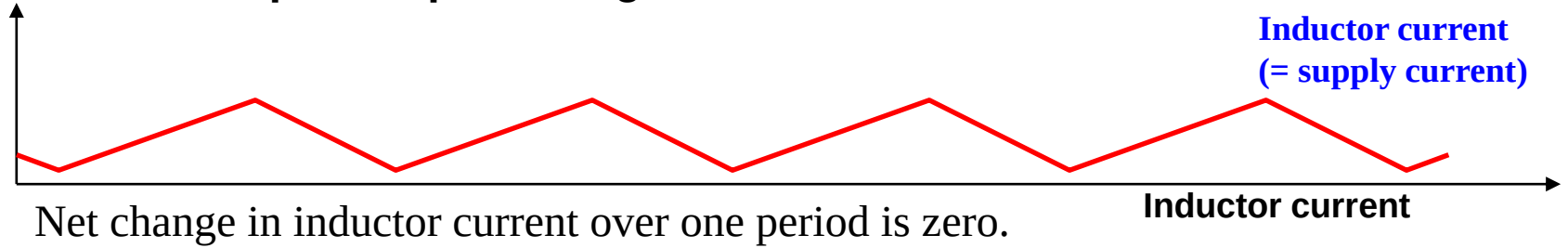


When switch is open:

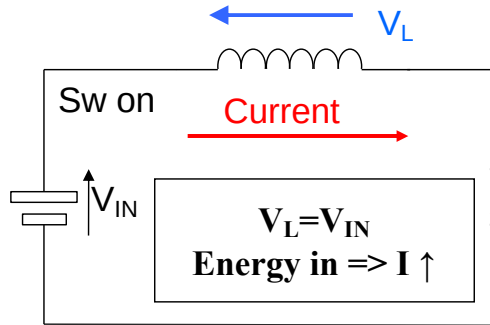
$$\frac{(\Delta i_L)_{switch\ open}}{T - t_{on}} = \frac{V_{IN} - V_{OUT}}{L} \Rightarrow (\Delta i_L)_{switch\ open} = \frac{V_{IN} - V_{OUT}}{L} \times (T - t_{on})$$



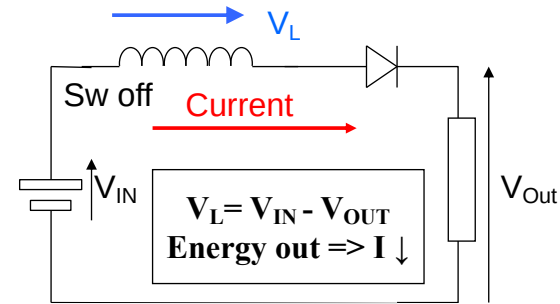
Boost converter Input/Output Voltage Relation



$$(\Delta i_L)_{\text{switch closed}} + (\Delta i_L)_{\text{switch open}} = 0.$$



$$\frac{V_{IN}}{L} \times t_{on} + \frac{V_{IN} - V_{OUT}}{L} \times (T - t_{on}) = 0 \Rightarrow$$



$$\frac{V_{IN}}{L} \times t_{on} + \frac{V_{IN}}{L} \times (T - t_{on}) - \frac{V_{OUT}}{L} \times (T - t_{on}) = 0$$

$$\frac{V_{OUT}}{L} \times (T - t_{on}) = \frac{V_{IN}}{L} \times (t_{on} + T - t_{on}) \Rightarrow$$

$$V_{OUT} \times (T - t_{on}) = V_{IN} \times (T)$$

$$V_{OUT} = V_{IN} \frac{T}{(T - t_{on})} \Rightarrow$$

$$V_{OUT} = V_{IN} \frac{\frac{T}{T}}{(\frac{T}{T} - \frac{t_{on}}{T})}$$

$$\text{Duty cycle of switching signal} = D = \frac{t_{on}}{T}$$

$$V_{OUT} = V_{IN} \frac{1}{(1 - D)}$$

Minimum and Maximum Inductor Current

Assuming lossless system

$$P_{IN} = P_{OUT} \quad \Rightarrow \quad V_{IN} I_{L,avg} = V_{OUT} I_{OUT} \quad \Rightarrow \quad V_{IN} I_{L,avg} = V_{OUT} \frac{V_{OUT}}{R}$$

$$V_{IN} I_{L,avg} = \frac{V_{OUT}^2}{R} \quad \Rightarrow \quad V_{IN} I_{L,avg} = \left(\frac{V_{IN}}{1-D} \right)^2 \times \frac{1}{R}$$

$$I_{L,avg} = \frac{V_{IN}}{(1-D)^2 R}$$

$$I_{L,max} = I_{L,avg} + \frac{(\Delta i_L)_{switch\ closed}}{2}$$

$$I_{L,min} = I_{L,avg} - \frac{(\Delta i_L)_{switch\ closed}}{2}$$

Note: Since change in inductor current is zero, you can take $(\Delta i_L)_{switch\ closed}$ or $(\Delta i_L)_{switch\ open}$ to calculate max and min current. You need to consider the sign of the current.

Boost converter Output Voltage Ripples

- Assuming that all the ripple current component of the diode current flows through the capacitor and its average value flows through the load resistor.

$$i_C = i_L - i_R$$

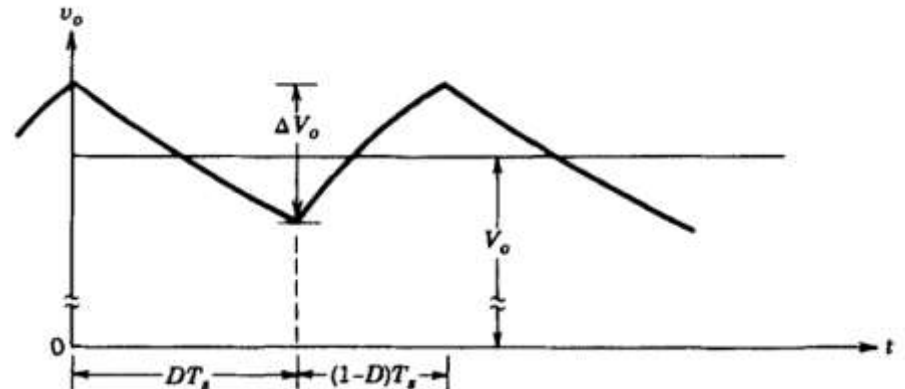
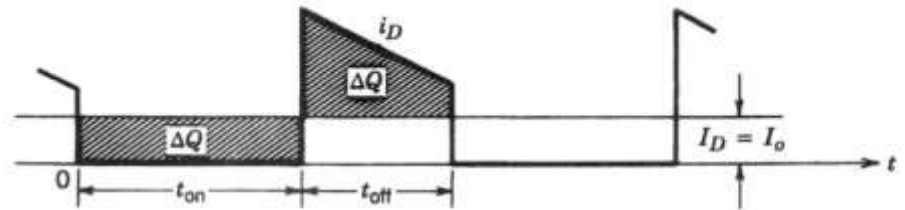
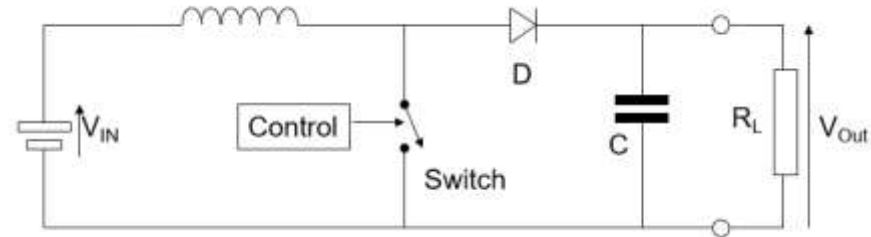
$$\Delta Q = C \Delta V_{OUT} \quad \Delta V_{OUT} = \frac{\Delta Q}{C}$$

$$\Delta V_{OUT} = \frac{I_o t_{on}}{C}$$

$$\Delta V_{OUT} = \frac{V_{OUT} t_{on}}{R C}$$

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{t_{on}}{R C}$$

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{DT}{R C}$$



Boost tutorial.

A boost converter has following parameters. Determine duty cycle, average, minimum and maximum inductor current and output voltage ripple.

$$V_{IN} = 12 \text{ V}$$

$$V_{OUT} = 30 \text{ V}$$

$$L = 120 \text{ } \mu\text{H}$$

$$C = 48 \text{ } \mu\text{F}$$

$$R = 50 \text{ ohm}$$

$$f = 50 \text{ KHz}$$

$$V_{OUT} = \frac{V_{IN}}{1 - D}$$

$$30 = \frac{12}{1 - D} \Rightarrow D = 0.6$$

Boost tutorial.

A boost converter has following parameters. Determine duty cycle, average, minimum and maximum inductor current and output voltage ripple.

$$\begin{aligned}V_{IN} &= 12 \text{ V} \\V_{OUT} &= 30 \text{ V} \\L &= 120 \mu\text{H} \\C &= 48 \mu\text{F} \\R &= 50 \text{ ohm} \\f &= 50 \text{ KHz}\end{aligned}$$

$$I_{L,avg} = \frac{V_{IN}}{(1-D)^2 R}$$

$$\Rightarrow I_{L,avg} = \frac{12}{(1-0.6)^2 50} = 1.5$$

$$(\Delta i_L)_{\text{switch closed}} = \frac{V_{IN}}{L} \times t_{on} = \frac{V_{IN}}{L} \times DT = \frac{V_{IN}}{L} \times \frac{D}{f}$$

$$(\Delta i_L)_{\text{switch closed}} = \frac{12}{120 \mu\text{H}} \times \frac{0.6}{25000} = 2.4 \text{ A}$$

$$I_{L,max} = I_{L,avg} + \frac{(\Delta i_L)_{\text{switch closed}}}{2}$$

$$I_{L,min} = I_{L,avg} - \frac{(\Delta i_L)_{\text{switch closed}}}{2}$$

$$I_{L,max} = 1.5 + 1.2 = 2.7 \text{ A}$$

$$I_{L,min} = 1.5 - 1.2 = 0.3 \text{ A}$$

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{DT}{RC} \Rightarrow \frac{\Delta V_{OUT}}{V_{OUT}} = \frac{D}{RCf}$$

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{0.6}{50 \times 48 \mu\text{F} \times 25000} = 0.01$$

Example.

A boost converter is required to provide a constant 24V / 100W output from an input voltage that varies between 9V and 15V. For a switching frequency of 20KHz and assuming 100% efficiency and operation in continuous mode, determine the following:

1. The required PWM duty cycle range to maintain a 24V DC output.
2. The average input current when the input voltage is 15V.
3. The value of the inductor required if the instantaneous current through the inductor at the end of the PWM period is zero. Calculate the value for both 9V and 15V input.

Solution:

1. For boost converter

$$V_{OUT} = \frac{V_{IN}}{(1-D)} \Rightarrow D = 1 - \frac{V_{IN}}{V_{OUT}} \quad \text{For } V_{IN} = 9V \quad D = 1 - \frac{9}{24} = 62.5\%$$
$$\text{For } V_{IN} = 15V \quad D = 1 - \frac{15}{24} = 37.5\%$$

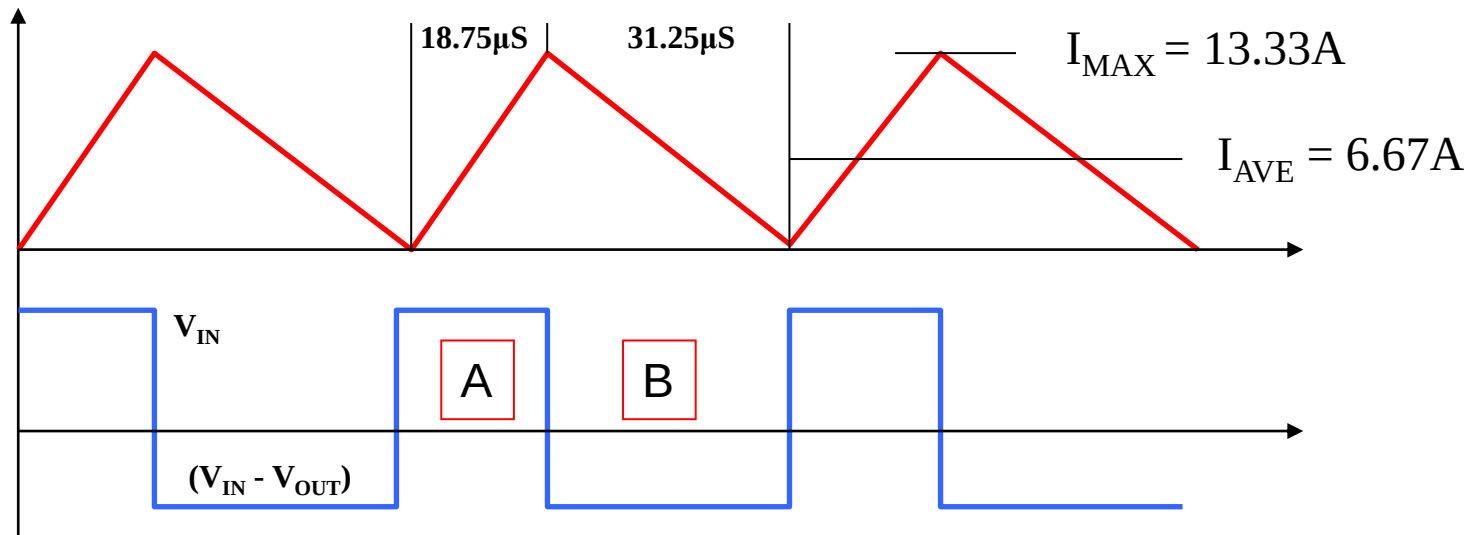
2. Assuming 100% efficiency:

$$V_{IN} I_{IN(MEAN)} = V_{OUT} I_{OUT(MEAN)} = 100W$$

$$\text{For } V_{IN} = 15V, I_{IN(MEAN)} = \frac{100W}{15V} = 6.67A$$

3. For a boost converter. the inductor current = input current. At 15V, $D=37.5\%$ and $I_{IN(AVE)}=6.67A$. We know that at the end of the PWM period the inductor current is zero, hence we can construct the current waveform:

$$I_{MAX} = 2 \times I_{AVE} = 13.33A$$



$$f = 20KHz, D = 37.5\% \Rightarrow T_{ON} = 18.75\mu s$$

$$V = L \frac{di}{dt} \Rightarrow L = V \frac{\Delta t}{\Delta i}$$

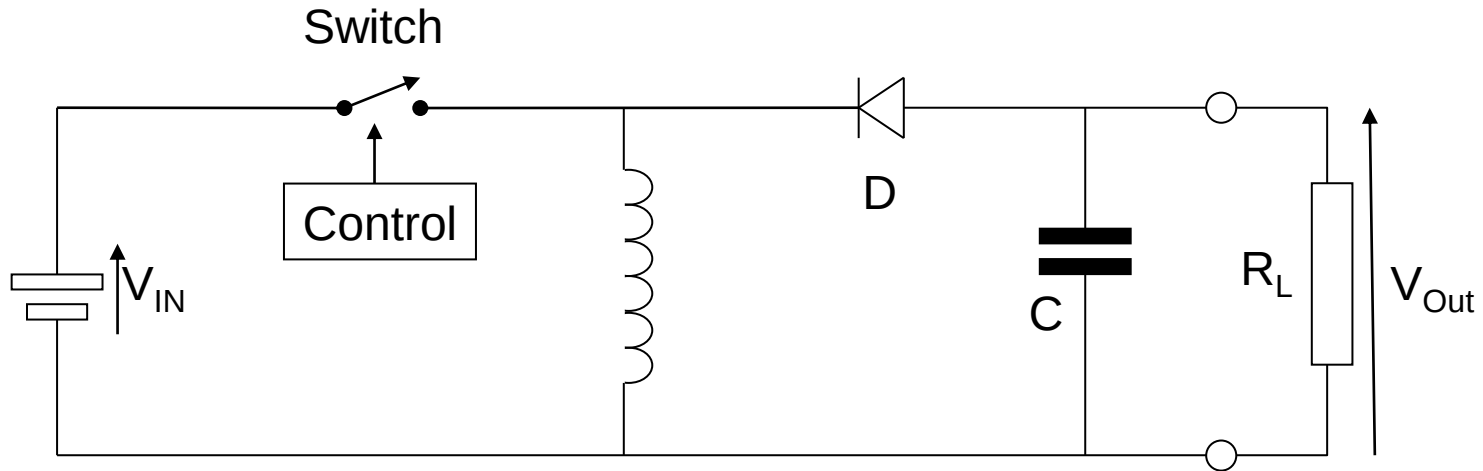
$$\text{For A: } L = \frac{15V \times 18.75\mu s}{13.33A} = 21\mu H$$

$$\text{For B: } L = \frac{-(24V - 15V) \times 31.25\mu s}{-13.33A} = 21\mu H$$

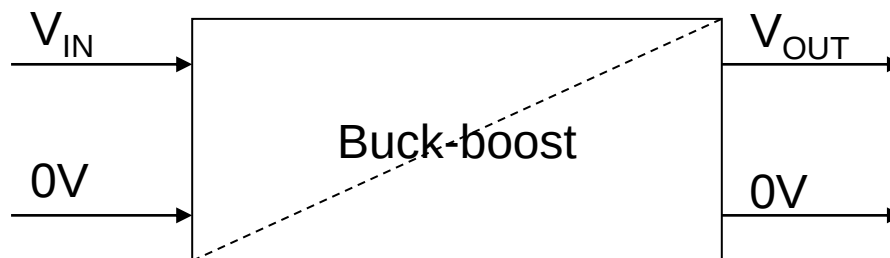
The Buck-Boost Converter

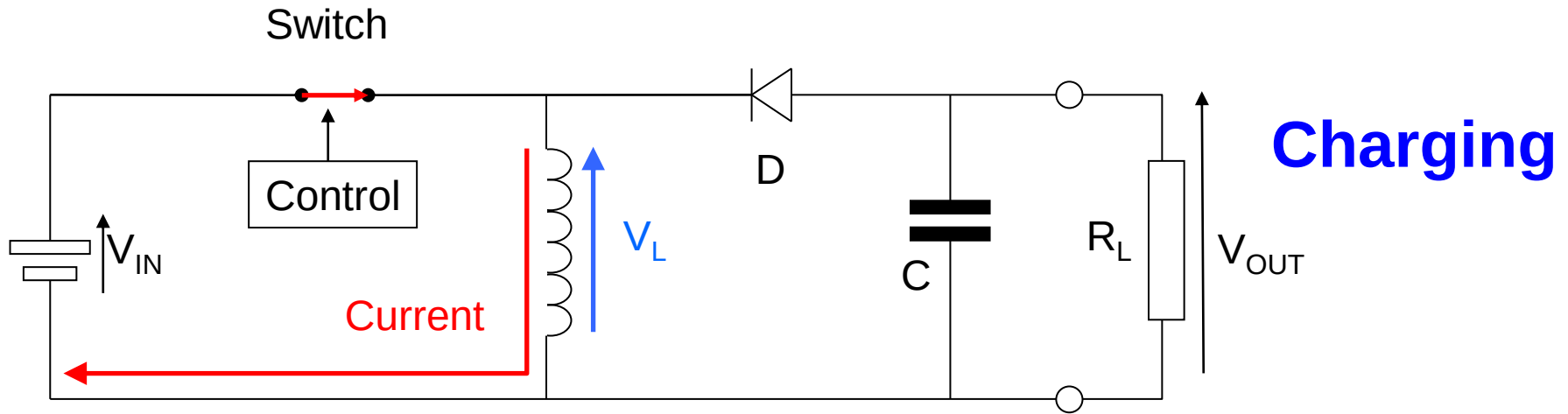
- How it works?
- Important waveforms
- Expression for output voltage
 - Based on average voltage of inductor
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The buck-boost converter ($V_{OUT} \leq V_{IN}$).



The Buck-Boost converter is able to produce output voltages that are higher or lower than the input voltage. Once again, we will consider the continuous mode system. Note that the output is negative with respect to the 0V side of the input voltage.





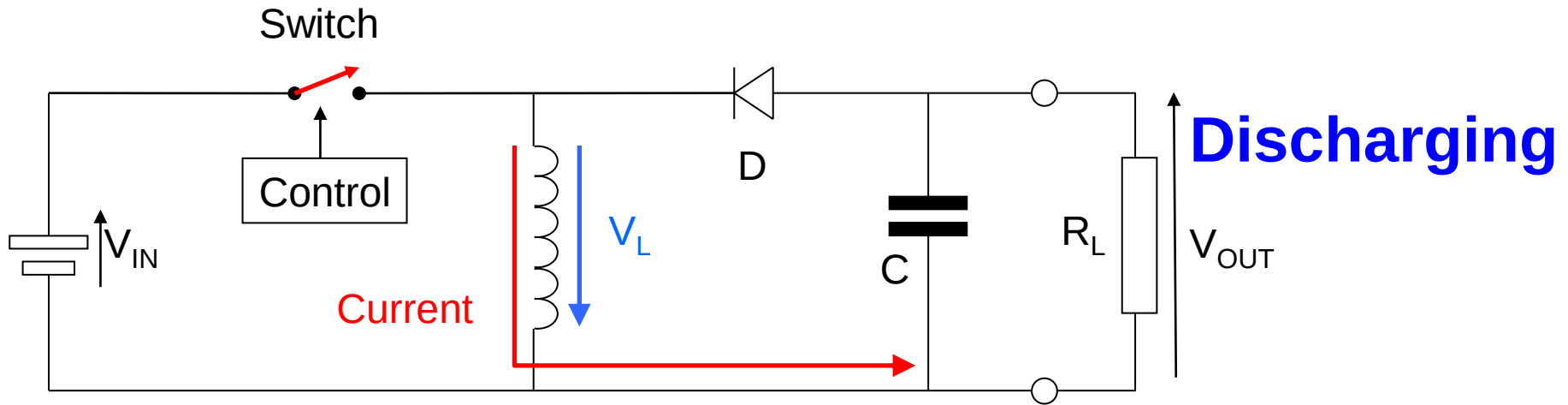
Before starting ...

... Notice that when the switch is open there is no connection between the input to the supply and its output. Hence, we can conclude that the supply is capable of delivering 0V output regardless of the input voltage for a 0% duty cycle on the switch.

With the switch closed, the input voltage is imposed across the inductor and the current through the inductor starts to rise. The diode is “off” since the cathode is more positive than the anode.

Hence for the “on” period, $V_L = V_{IN}$.

Note that with the diode reverse biased, the only route for the current to take that completes the circuit is back through the supply to the converter.



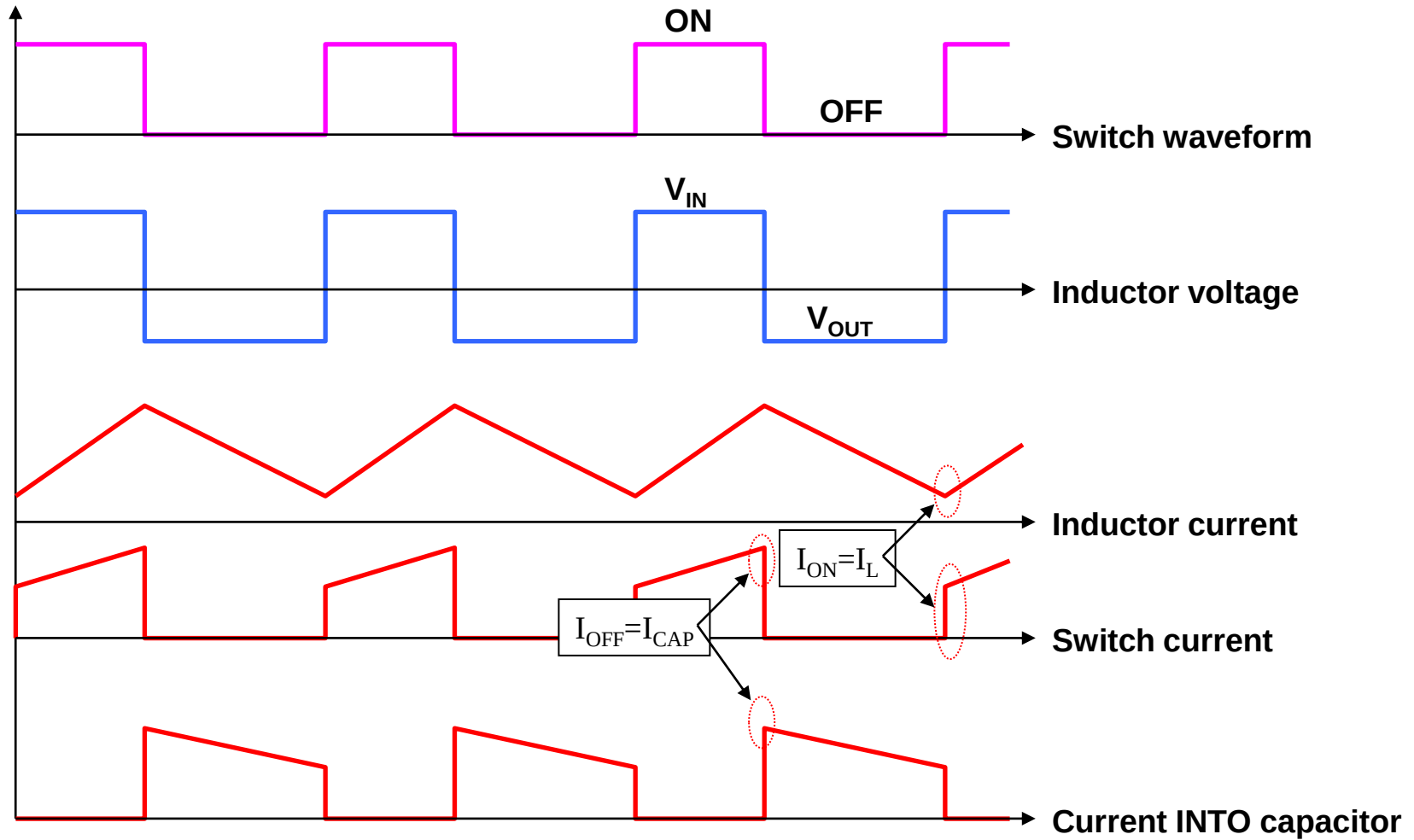
When the switch is opened the voltage on the inductor reverses polarity, the diode starts to conduct and hence the inductor current flows into the capacitor.

During this “off” period, $V_L = V_{OUT}$ and this is maintained as the current in the inductor falls to its value at the end of the PWM period.

Note that with the switch opened there is no return path for current through the input to the converter and hence the only route the inductor current can follow is into the capacitor and load.

Also note that the capacitor will supply current to the load when there is no current contribution from the inductor.

Waveforms for the buck-boost converter



Buck-boost converter Input/Output Voltage Relation

$$\int_0^T V_L dt = 0$$

$$\int_0^{t_{on}} V_{IN} dt + \int_{t_{on}}^T V_{OUT} dt = 0$$

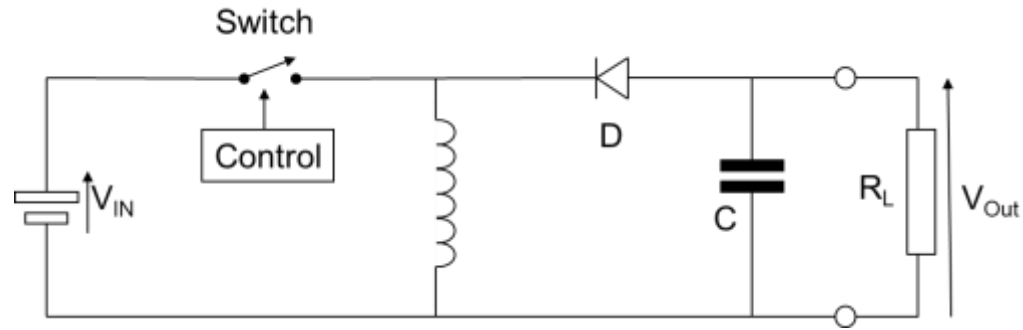
$$V_{IN}t_{on} + V_{OUT}T - V_{OUT}t_{on} = 0$$

$$V_{IN}t_{on} = -V_{OUT}T + V_{OUT}t_{on}$$

$$V_{IN}DT = -V_{OUT}T + V_{OUT}DT$$

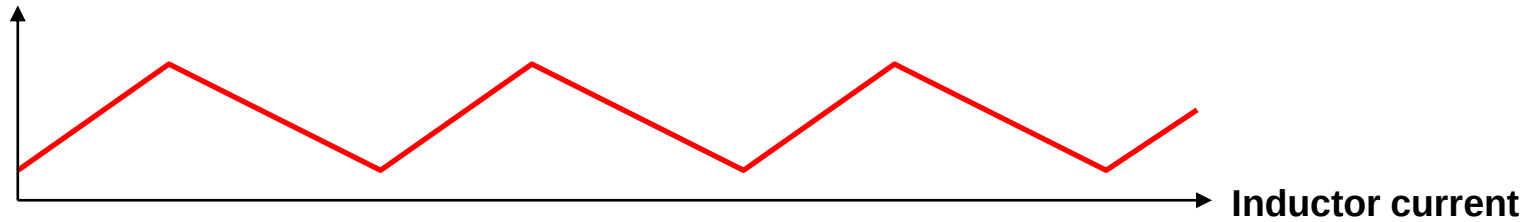
$$V_{IN}D = -V_{OUT}(1 - D)$$

$$V_{OUT} = \frac{-V_{IN}D}{1 - D}$$



Polarity is opposite than the input

Buck-boost converter change in inductor current



Computing the change in current using following expression:

$$v_L = L \frac{di_L}{dt} \Rightarrow \frac{di_L}{dt} = \frac{v_L}{L}$$

When switch is closed:

$$\frac{(\Delta i_L)_{switch\ closed}}{t_{on}} = \frac{V_{IN}}{L} \Rightarrow (\Delta i_L)_{switch\ closed} = \frac{V_{IN}}{L} \times t_{on}$$

When switch is open:

$$\frac{(\Delta i_L)_{switch\ open}}{T - t_{on}} = \frac{V_{OUT}}{L} \Rightarrow (\Delta i_L)_{switch\ open} = \frac{V_{OUT}}{L} \times (T - t_{on})$$

Minimum and Maximum Inductor Current

Assuming lossless system

$$P_{IN} = P_{OUT} \quad \Rightarrow \quad V_{IN} I_{IN} = V_{OUT} I_{OUT} \quad \Rightarrow \quad V_{IN} I_{L,avg} D = V_{OUT} \frac{V_{OUT}}{R}$$

$$V_{IN} I_{L,avg} D = \frac{V_{OUT}^2}{R} \quad \Rightarrow \quad I_{L,avg} = \frac{(V_{OUT})^2}{V_{IN} D} \times \frac{1}{R}$$

$$I_{L,avg} = \left(V_{IN} \frac{D}{(1-D)} \right)^2 \times \frac{1}{V_{IN} D R}$$

$$I_{L,avg} = \frac{V_{IN} D}{(1-D)^2 R}$$

$$I_{L,max} = I_{L,avg} + \frac{(\Delta i_L)_{switch\ closed}}{2} \quad \quad I_{L,min} = I_{L,avg} - \frac{(\Delta i_L)_{switch\ closed}}{2}$$

Note: Since change in inductor current is zero, you can take $(\Delta i_L)_{switch\ closed}$ or $(\Delta i_L)_{switch\ open}$ to calculate max and min current. You need to consider the sign of the current.

Buck-Boost converter Output Voltage Ripples

- Assuming that all the ripple current component of the diode current flows through the capacitor and its average value flows through the load resistor.

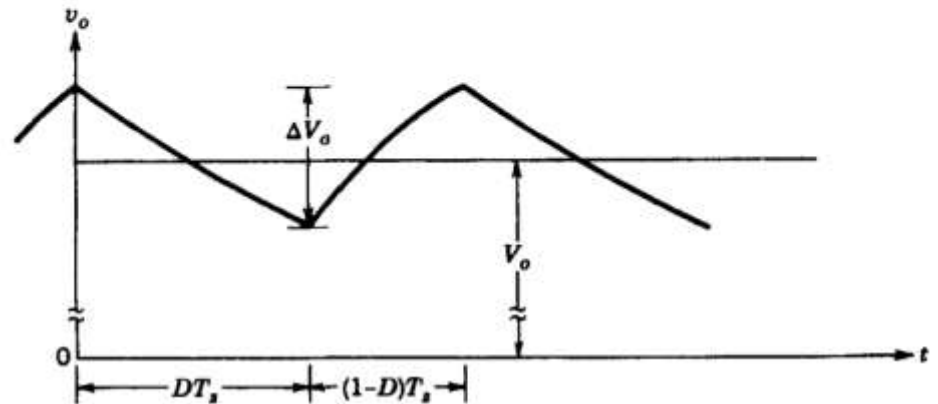
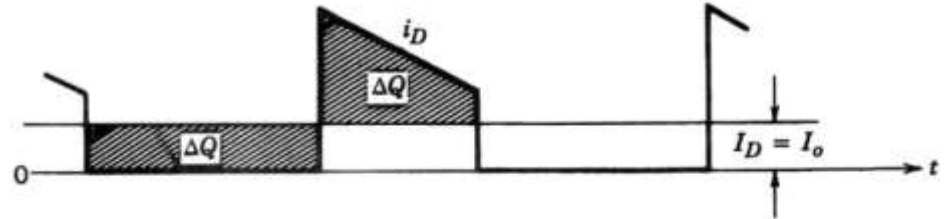
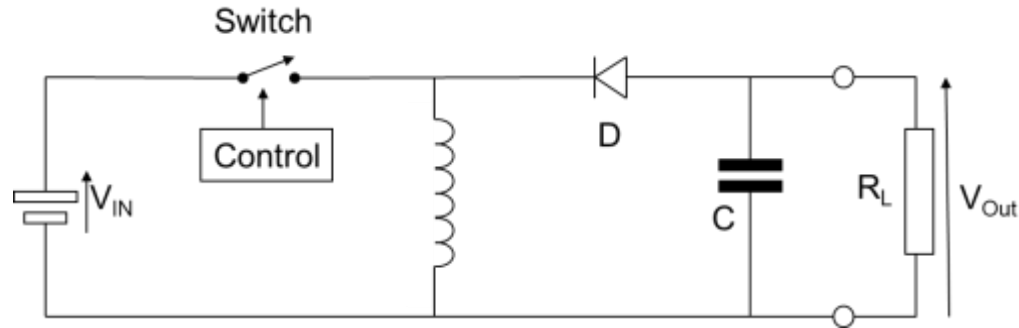
$$\Delta Q = C \Delta V_{OUT} \quad \Delta V_{OUT} = \frac{\Delta Q}{C}$$

$$\Delta V_{OUT} = \frac{I_o t_{on}}{C}$$

$$\Delta V_{OUT} = \frac{V_{OUT} t_{on}}{R C}$$

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{t_{on}}{R C}$$

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{DT}{R C}$$



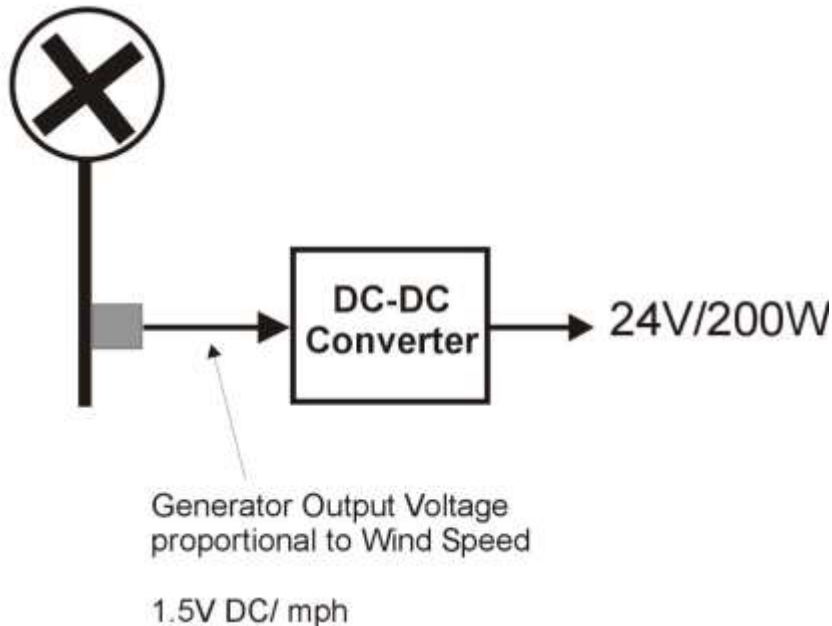
Question.

A small wind turbine generates 1.5V DC per MPH of wind speed.

For operation over the range 8 MPH to 32 MPH, determine the following:

1. The type of converter to be used;
2. The required range of duty cycle for these wind speeds;
3. The average input current to the converter at 32MPH;
4. The peak input current @ 32 MPH assuming operation at the boundary condition.

Wind Generator



Solution.

For a wind speed of 8MPH, the turbine output voltage is $8 \times 1.5 = 12\text{V}$. At a wind speed of 32 MPH the output voltage is 48V. Hence use a buck-boost converter since V_{IN} can be less than or greater than the output voltage.

For the BB,

$$V_{OUT} = -V_{IN} \frac{D}{(1 - D)}$$

Neglecting the sign as it only shows the polarity with respect to input voltage.

$$D = \frac{V_{OUT}}{(V_{IN} + V_{OUT})}$$

For 8MPH, $D = 24 / (12 + 24) = 66.6\%$

For 32 MPH, $D = 24 / (48 + 24) = 33.3\%$

Converter summary

For all converters, assume $V_{IN} I_{IN(Ave)} = V_{OUT} I_{OUT(Ave)}$

Type	Voltage Equation	Current Equation
Buck ($V_{OUT} < V_{IN}$)	$V_{OUT} = DV_{IN}$	$I_{OUT(Ave)} = \frac{I_{IN(Avg.)}}{D}$
Boost ($V_{OUT} > V_{IN}$)	$V_{OUT} = \frac{V_{IN}}{(1 - D)}$	$I_{OUT} = (1 - D)I_{IN(Avg.)}$
Buck-boost ($V_{OUT} <> V_{IN}$)	$V_{OUT} = -V_{IN} \frac{D}{(1 - D)}$	$I_{OUT(Ave)} = -I_{IN(Avg)} \frac{1 - D}{D}$

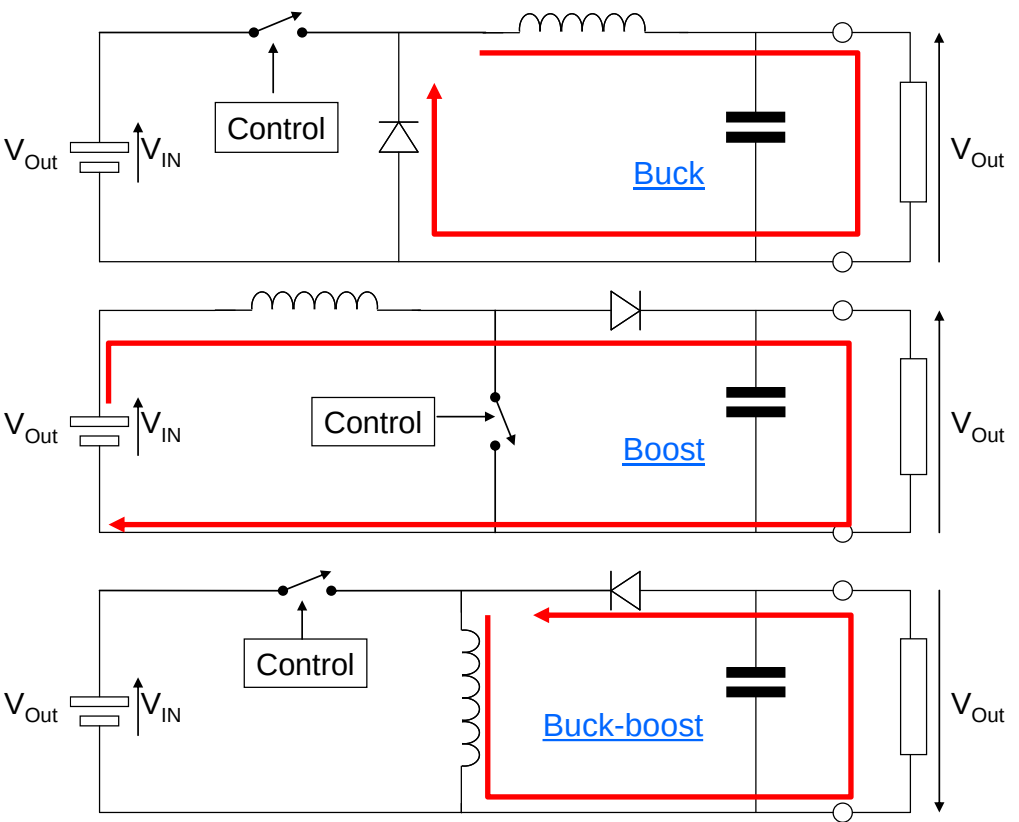
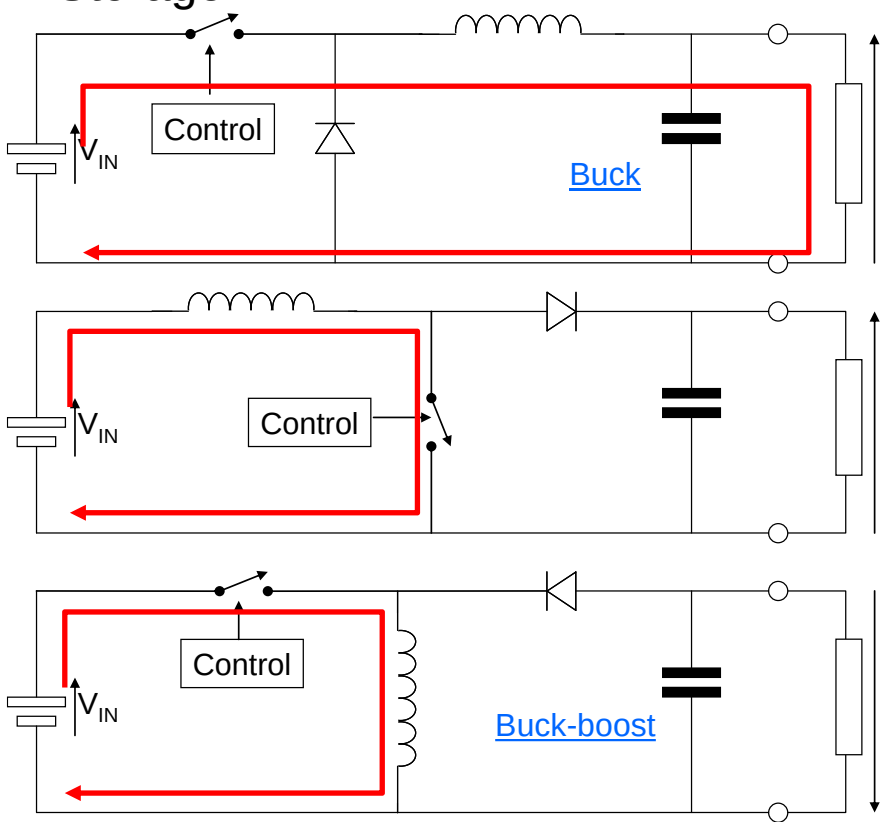
The following diagrams summarise the energy flow in the circuit for each converter type during the storage and release phases of operation.

Charging the Inductor

Discharging the Inductor

Storage

Release



Switch ON

Switch OFF

What have we learned so far?

- DC-DC Converters
- Switched Mode Power Supplies (Converters)
 - Buck Converter
 - Boost Converter
 - Buck-Boost Converter